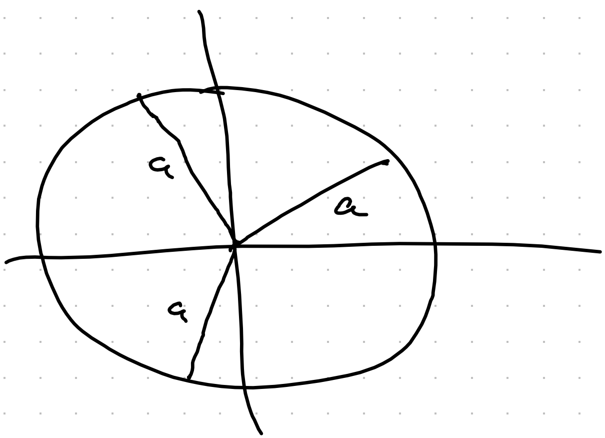
Functions in polar coordinates — 7.3

Functions of the form $r = f(\theta)$

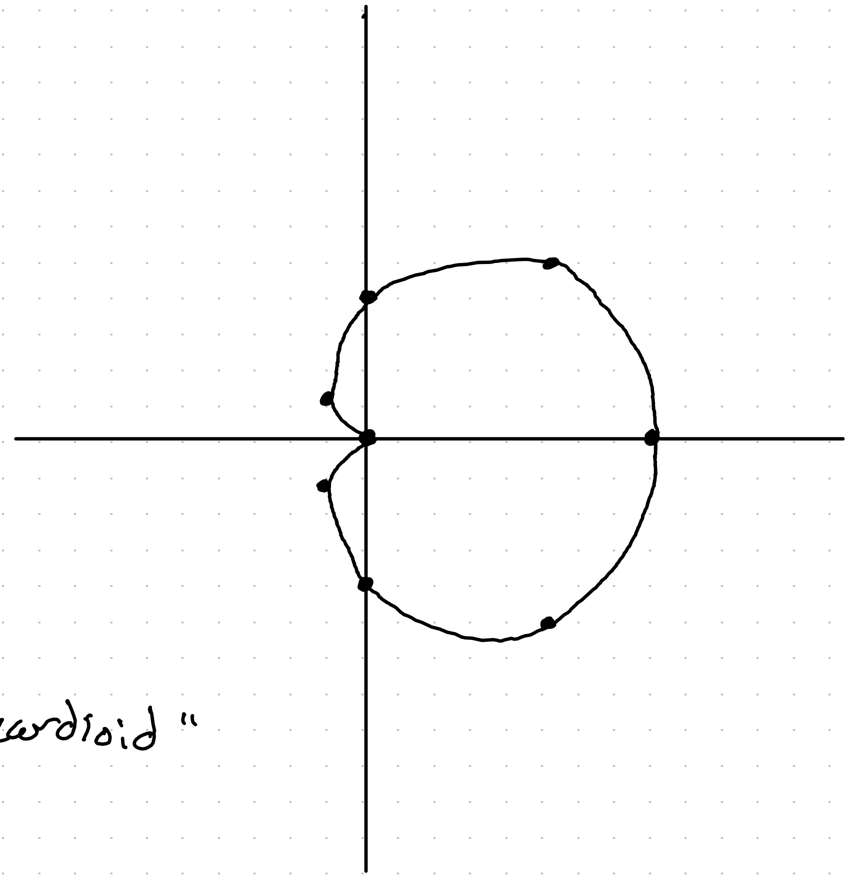
What do constant functions look like?

$r = a$ is a circle of radius a



Example: Sketch the curve $r = 1 + \cos(\theta)$.

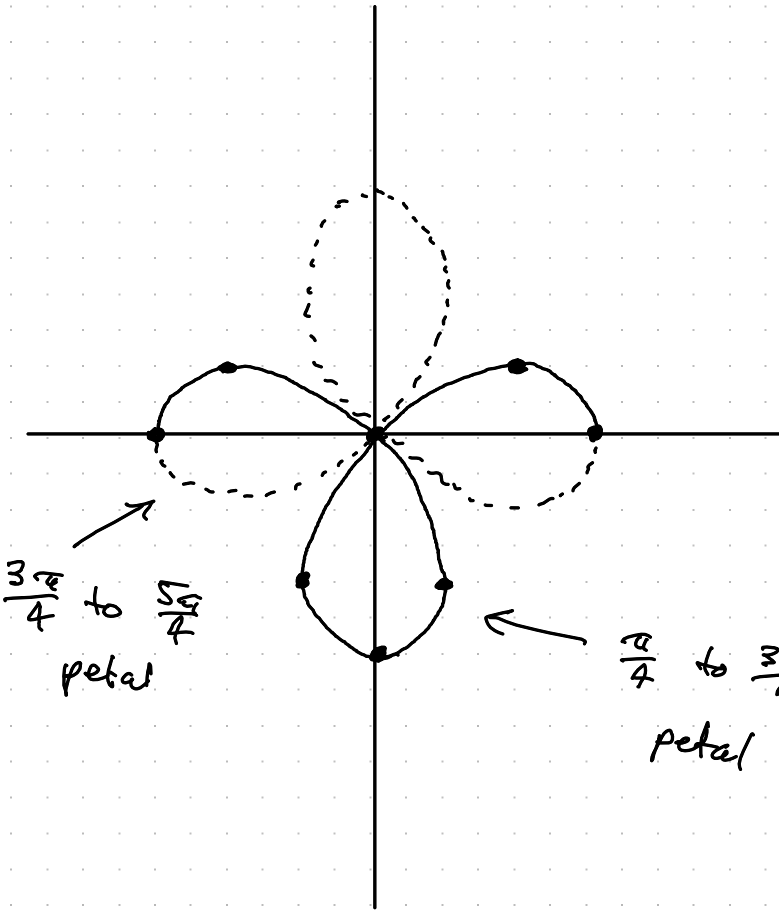
θ	r
0	2
$\frac{\pi}{4}$	$1 + \frac{\sqrt{2}}{2}$
$\frac{\pi}{2}$	1
$\frac{3\pi}{4}$	$1 - \frac{\sqrt{2}}{2}$
π	0
$\frac{5\pi}{4}$	$1 - \frac{\sqrt{2}}{2}$
$\frac{3\pi}{2}$	1
$\frac{7\pi}{4}$	$1 + \frac{\sqrt{2}}{2}$
2π	2



"cardioid"

Example: Sketch the curve $r = \cos(2\theta)$.

θ	r
0	1
$\frac{\pi}{8}$	$\frac{\sqrt{2}}{2}$
$\frac{\pi}{4}$	0
$\frac{3\pi}{8}$	$-\frac{\sqrt{2}}{2}$
$\frac{\pi}{2}$	-1
$\frac{5\pi}{8}$	$-\frac{\sqrt{2}}{2}$
$\frac{3\pi}{4}$	0
$\frac{7\pi}{8}$	$\frac{\sqrt{2}}{2}$
π	1

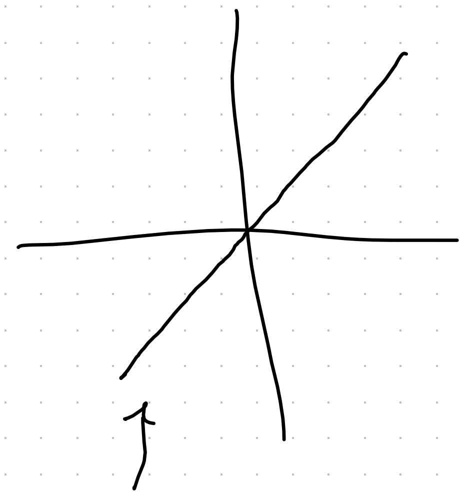


Other common shapes:

$r = 2a\sin(\theta)$ — circle of radius a centered $(0,a)$

$r = 2a\cos(\theta)$ — " " $(a,0)$

$\theta = a$ gives a line through the origin



$\theta = \frac{\pi}{4}$

$r = m\theta + b$ gives a spiral